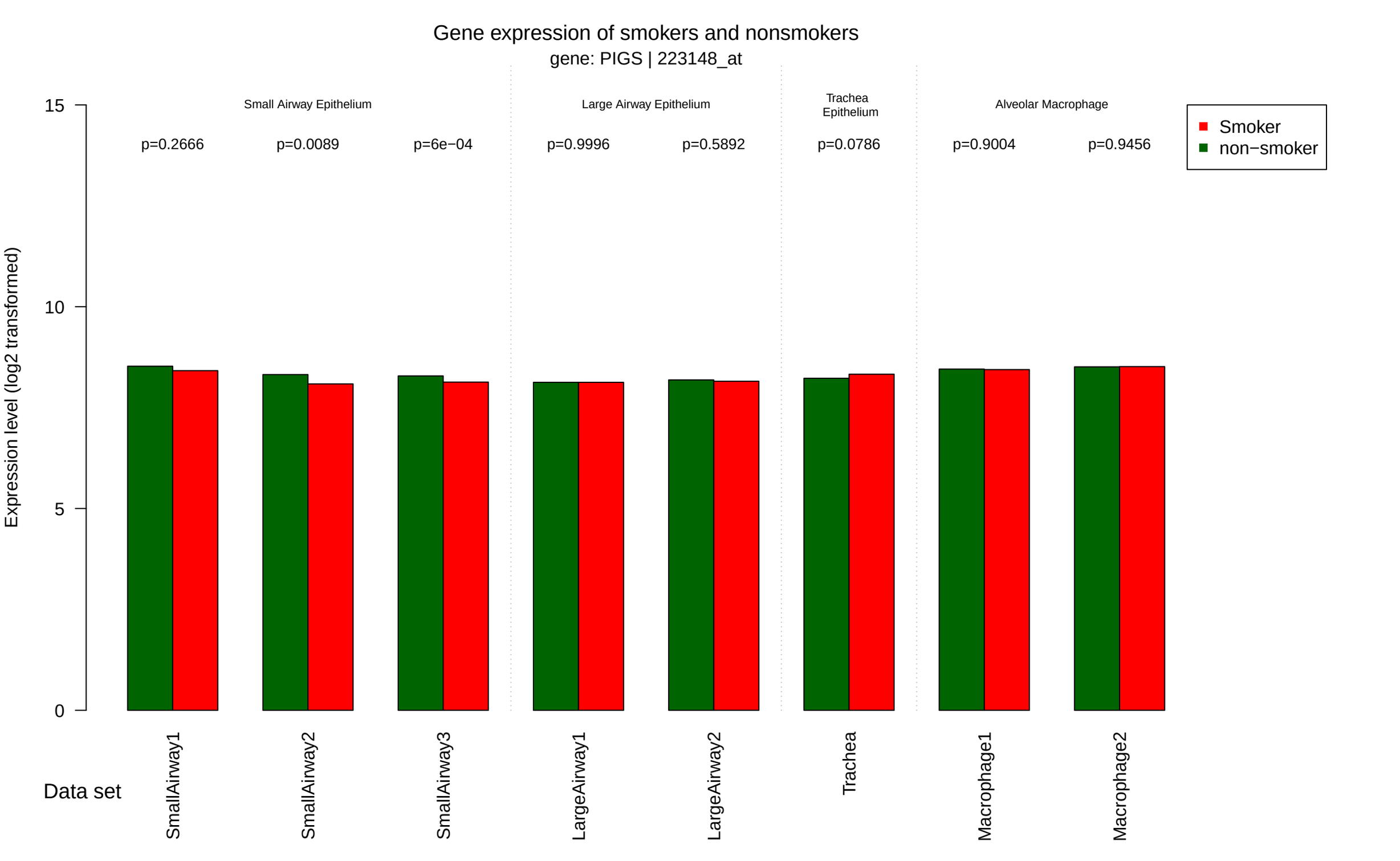
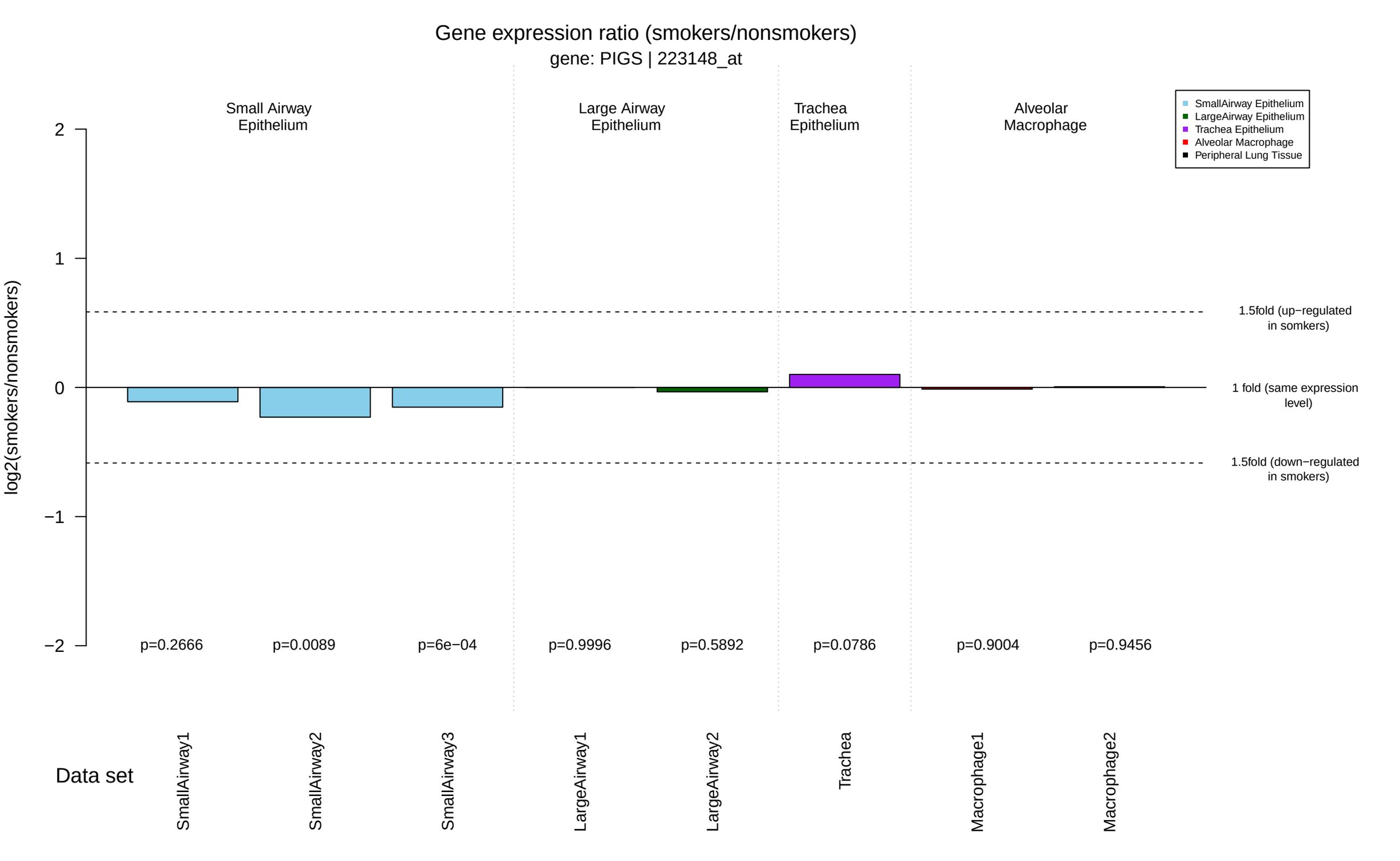
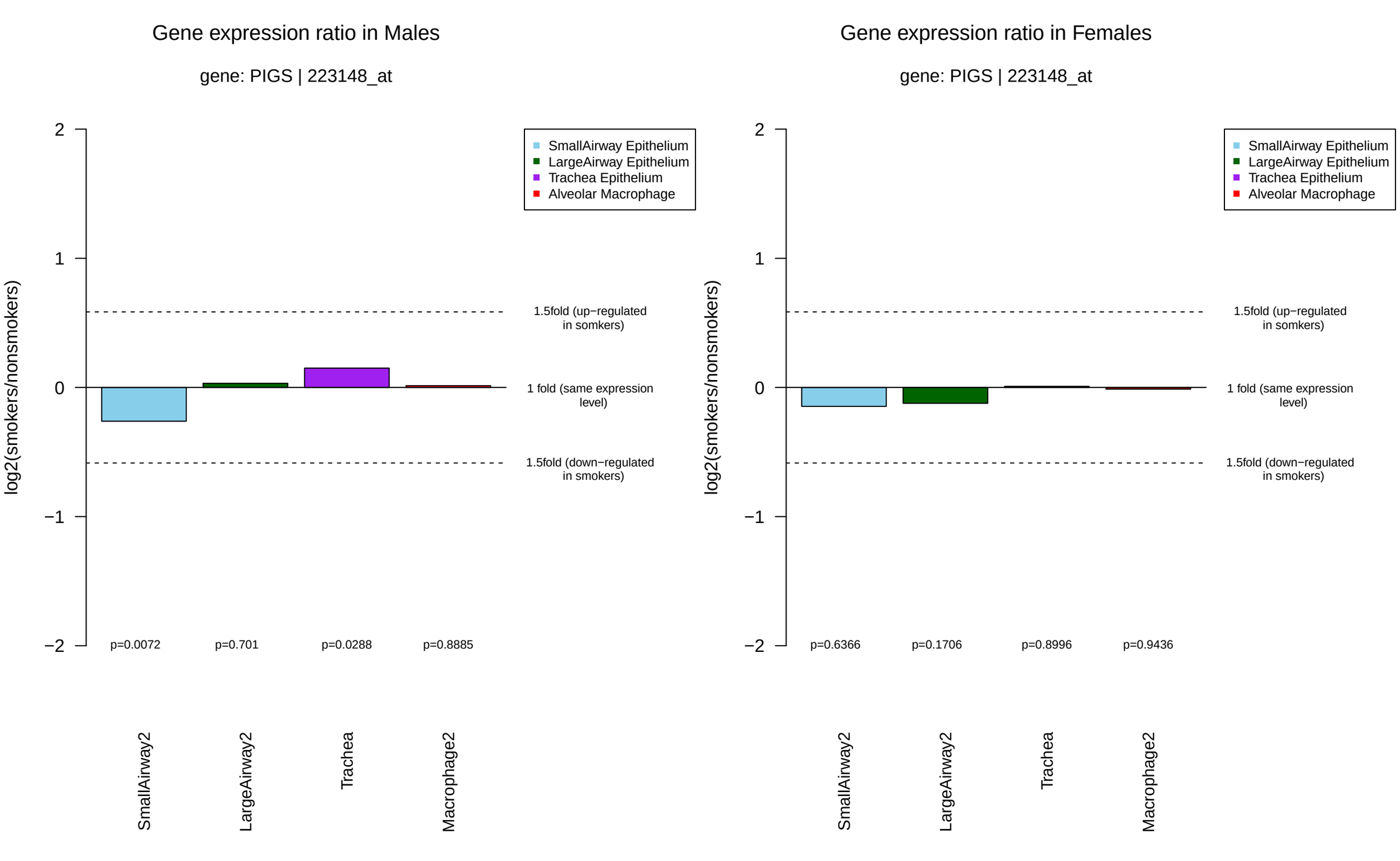
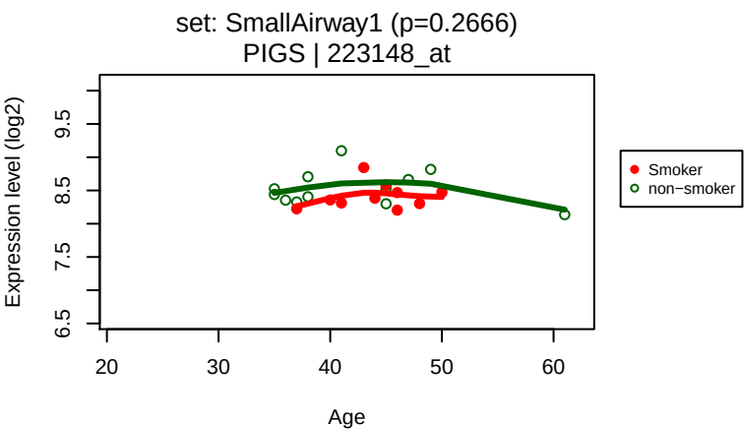
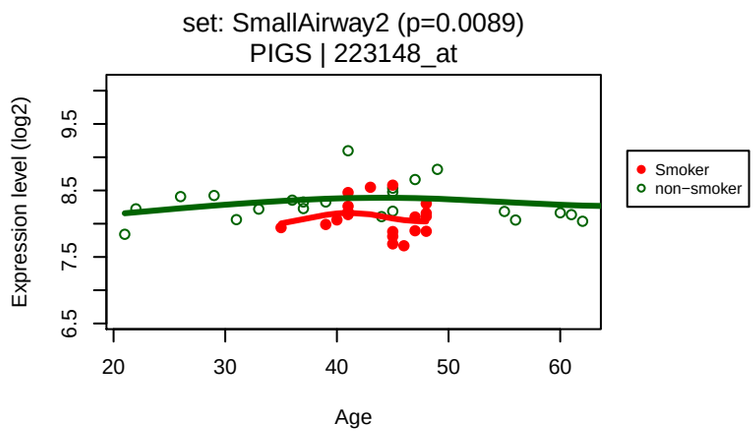
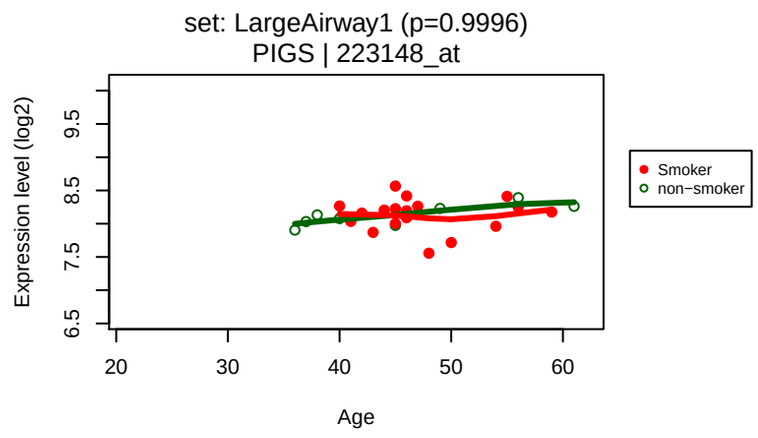

Gene expression of smokers and nonsmokers
gene: PIGS | 223148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2666
p=0.0089
p=6e−04
p=0.9996
p=0.5892
p=0.0786
p=0.9004
p=0.9456
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

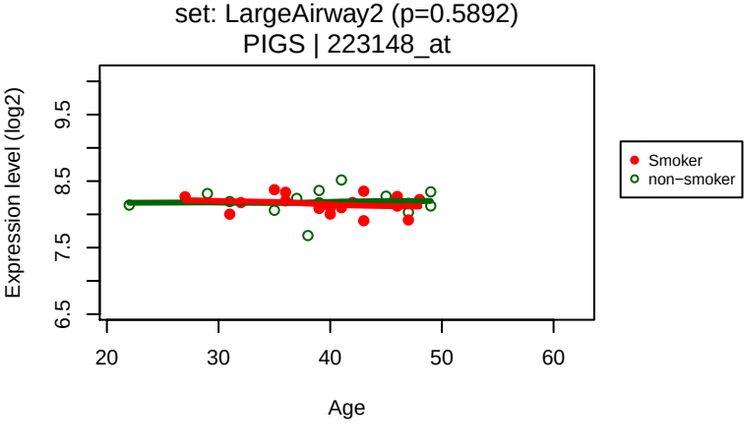

Gene expression ratio (smokers/nonsmokers)
gene: PIGS | 223148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2666
p=0.0089
p=6e−04
p=0.9996
p=0.5892
p=0.0786
p=0.9004
p=0.9456
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

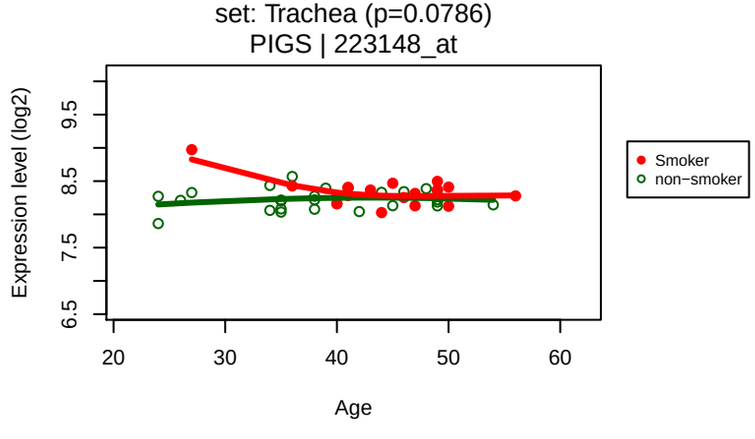

Gene expression ratio in Males
gene: PIGS | 223148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0072
p=0.701
p=0.0288
p=0.8885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIGS | 223148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6366
p=0.1706
p=0.8996
p=0.9436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

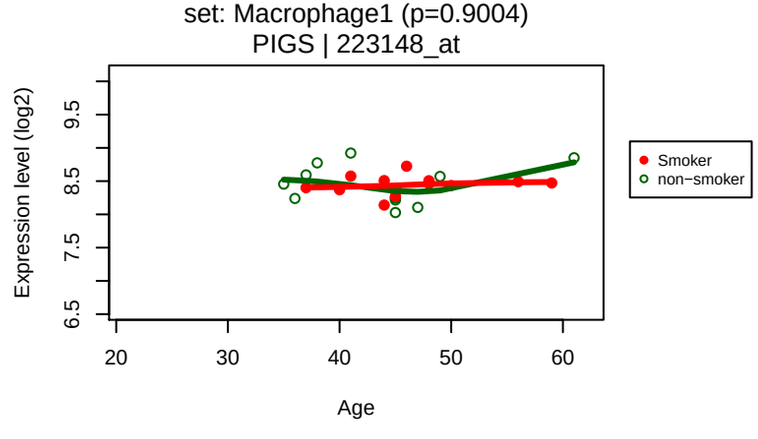

set: SmallAirway1 (p=0.2666)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

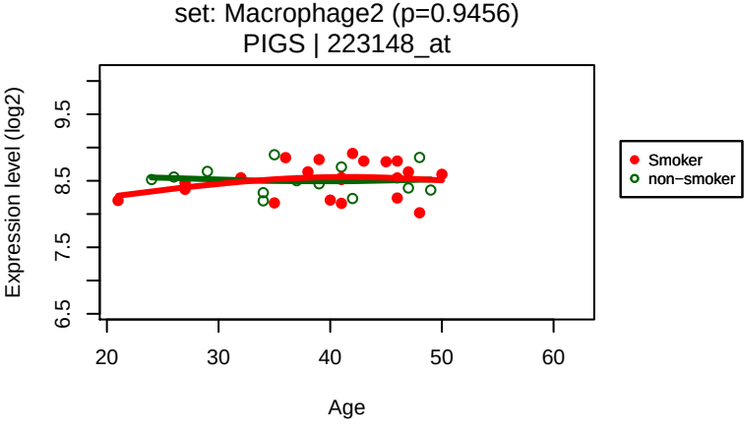

set: SmallAirway2 (p=0.0089)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

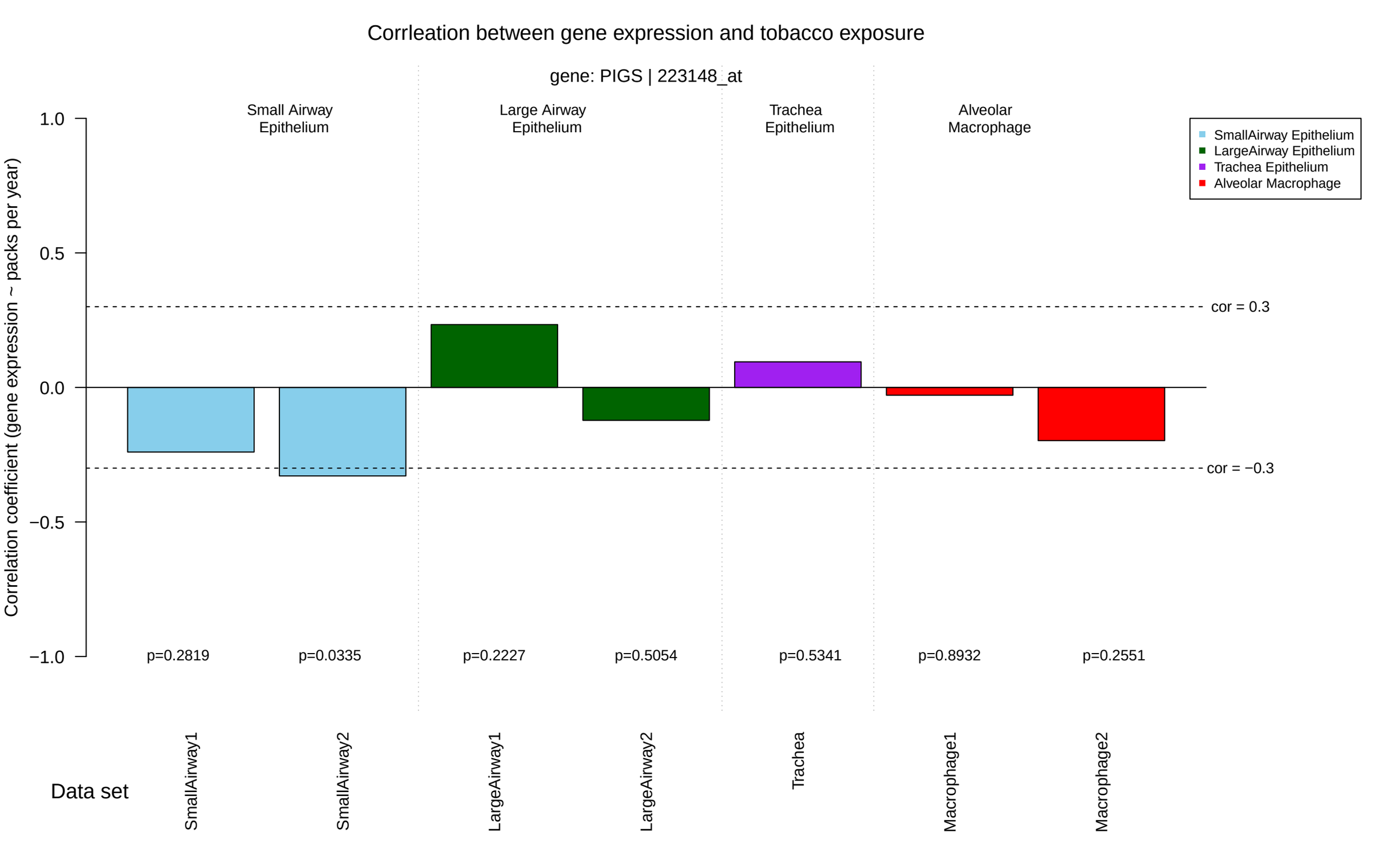

set: LargeAirway1 (p=0.9996)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5892)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0786)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9004)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9456)
PIGS | 223148_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIGS | 223148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2819
p=0.0335
p=0.2227
p=0.5054
p=0.5341
p=0.8932
p=0.2551
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2